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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fourth Semester B.E. (ISE) (Credit System) Degree Examinations

Supplementary Examinations – December 2025

21IS403 / 20IS405 – DIGITAL SYSTEM DESIGN

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain how NAND and NOR gates are considered universal gates with the help of logic gate constructions.	8	L*2	1	1
b)	Perform the binary subtraction using 2's complement method.				
i.	$(11010010)_2 - (01011110)_2$	6	L3	1	2
ii.	$(00110)_2 + (11010)_2$	6	L3	1	2
c)	Prove De-Morgan's theorems using the truth table approach.	6	L3	1	2
2. a)	Using K-map simplify the following function:				
i.	$F(a,b,c) = \sum m(0,1,2,3,7)$				
ii.	$F(a,b,c,d) = \sum m(0,1,3,4,5,6,9,11,12,13,14)$	8	L3	2	2
b)	Perform the conversion for the following and show each step clearly.				
i.	$1A2_{(16)} = (?)_{10}$				
ii.	$45.25_{(10)} = (?)_2$				
iii.	$12.6532_{(10)} = (?)_{16}$	6	L3	1	2
c)	Construct verilog code and logic circuit for the expression.				
i.	$y = AB + CD$	6	L3	2	2
ii.	$y = (A+B)(C+D)$	6	L3	2	2
3. a)	Simplify the following using Boolean algebra: $(A + B' + C')(A + B' + C)(A + B + C')$	6	L3	1	2
b)	A system receives three inputs and generates a one-bit output based on the even number of ones present in the input. If the inputs are completely zeros or if it has an odd number of ones, the output will be zero and if the input has an even number of ones then the output will be set to one. Write the truth table for such a system and represent the simplified function using the logic diagram.	6	L3	2	2
c)	Simplify the following expression using Quine McClusky Method. $F = \sum m(2,6,8,9,10,11,14,15)$.	8	L3	2	2
Unit – II					
4. a)	Implement $f(a,b,c) = \sum m(1,4,5,7)$ using				
i)	8:1 MUX	6	L3	3	2
ii)	4:1 MUX	8	L2	3	1
b)	Explain half adder and full adder with diagrams	6	L2	3	1
c)	Explain the working of 8 to 3 line encoder.				
5. a)	Implement the given function using 3 to 8 decoder with minimum number of gate inputs.				
i)	$F1(a,b,c) = \sum m(1,2,6)$				
ii)	$F2(a,b,c) = \sum m(0,3,5,6,7)$				
iii)	$F3(a,b,c) = \prod m(1,2,3,5,6,7)$	6	L3	3	2
b)	Design a circuit for one bit magnitude comparator along with truth table and equations.	6	L2	3	1
c)	Explain the operation of master slave JK flip-flop with truth table and timing diagram.	8	L2	4	1

P.T.O.

6. a) Explain with an example the working of 4-bit binary parallel adder/subtractor with relevant block diagram.

10 L2 3 1

- b) Convert:

- i) JK flip-flop to D flip-flop
ii) JK flip-flop to T flip-flop

10 L3 4 2

Unit – III

7. a) What is shift register? With the help of a suitable example explain the following operations:

- i) SIPO
ii) PIPO

10 L2 5 1

- b) Write characteristic equations and state transition diagram for D flip-flop and JK flip-flop.

10 L2 5 1

8. a) What is ripple counter? Explain the working of a 3-bit ripple-up counter using the positive edge-triggered T flip-flops with a relevant logic diagram and the count sequence.

10 L2 5 1

- b) Briefly explain the applications of shift registers.

10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
